

EE-312L Signals and Systems Lab  
Laboratory Session 11  
**To understand and demonstrate the  
Discrete-Time Fourier Transform (DTFT)  
and Discrete Fourier Transform (DFT) using  
MATLAB**

December 19, 2024

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## 1 Objective

The objective of this lab session is to understand and demonstrate continuous-Time Fourier transform (DTFT) and discrete Fourier transform (DFT) using MATLAB.

## 2 Discrete-Time Fourier Transform (DTFT)

A DT signal  $x[n]$  and its frequency domain, Fourier transform signal,  $X(e^{j\omega})$ , are related by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1)$$

where  $\omega = 2\pi f$ ,  $\omega$  is measured in  $rad/s$  and  $f$  is measured in  $Hz$ . The  $X(j\omega)$  in (1), is referred to as **Analysis Equation**.

To find the inverse DTFT, we use the **Synthesis Equation**, given in (1)

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega. \quad (2)$$

### 3 Discrete-Time Fourier Transform Using MATLAB

#### Example 01:

Compute the DTFT  $X(e^{j\omega})$  of the signal  $x[n] = 0.8^n$ ,  $0 \leq n \leq 20$  and plot  $X(e^{j\omega})$  over the frequency intervals  $-\pi \leq \omega \leq \pi$  and  $-5\pi \leq \omega \leq 5\pi$  [PV10].

```
1
2 clc;
3 clear all;
4 close all;
5
6 w = sym('w')
7 n = 0:2;
8 x = 0.8.^n;
9 X = sum(x.*exp(-j*w*n));
10
11 figure(1)
12 h1 = ezplot(abs(X),[-pi pi])
13 title('Magnitude of DTFT')
14 ylim([0 3])
15 set(h1, 'Color', 'r', 'LineWidth',2) ;
16
17
18 ##figure(2)
19 ##w1 =-pi:.01:pi;
20 ##XX = subs(X,w,w1)
21 ##figure(2)
22 ##plot(w1,angle(XX));
23 ##xlim([-pi pi])
24 ##title('Phase of DTFT')
25
26 figure(3)
27 h2 = ezplot(abs(X),[-5*pi 5*pi]);
28 title('Magnitude of DTFT in 5 periods')
29 ylim([0 3])
30 set(h2, 'Color', 'r', 'LineWidth',2) ;
31
32
33 ##w2 = -5*pi:.01:5*pi;
34 ##XX = subs(X,w,w2);
35 ##figure(4)
36 ##plot(w1,angle(XX));
37 ##xlim([-5*pi 5*pi])
38 ##title(' Phase of DTFT in 5 periods')
```

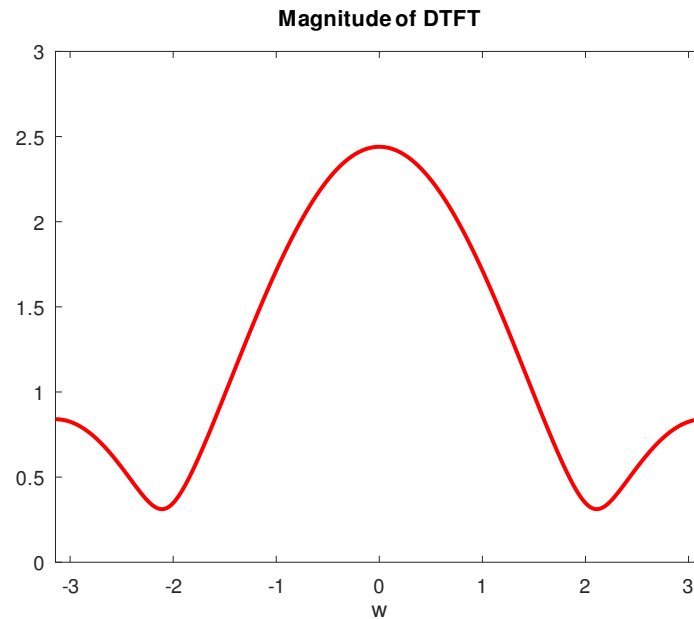


Figure 1: DTFT magnitude over one period

## 4 Discrete Fourier Transform [PV10]

To implement DTFT

- Obtain samples  $\omega_k$  from the frequency  $\omega$ .
- This leads to a second type of Fourier transform appropriate for discrete-time signals, the discrete Fourier transform (DFT).
- The  $N$ -point DFT of a discrete-time signal  $x[n]$  is denoted by  $X_k$  or  $X(\omega_k)$  or  $X(k)$
- It is defined in the discrete-time interval  $0 \leq n \leq N - 1$ , as follows

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi nk}{N}} \quad (3)$$

### Example 02:

Compute the DFT  $X_k$  of the sequence  $x[n] = [1, 2, 2, 1]$ ,  $0 \leq n \leq 3$ . Plot the magnitude, the phase, the real part, and the imaginary part of  $X_k$ .

```

1
2 clc;
3 clear all;
4 close all;
5
6 x = [1 2 2 1];
```

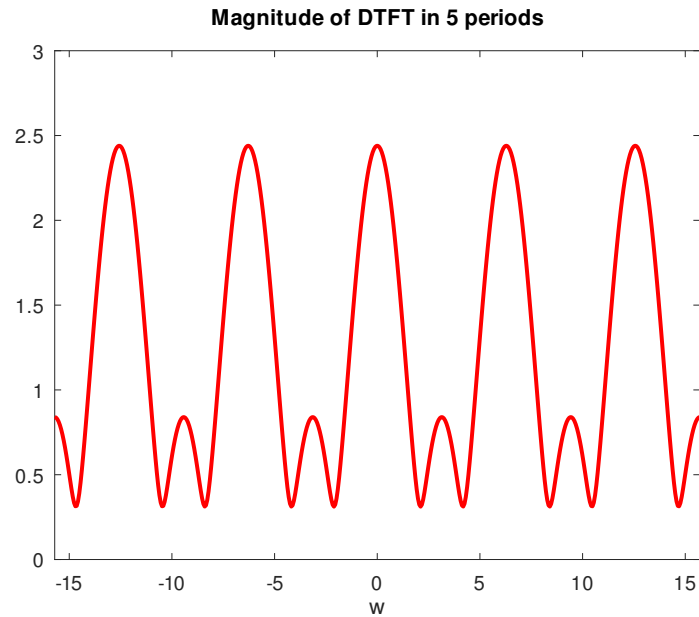


Figure 2: DTFT magnitude over five periods

```

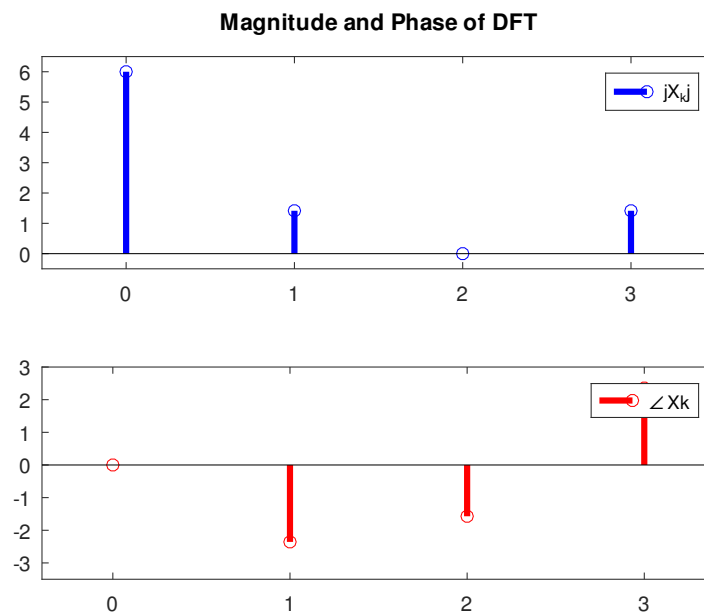
7 N = length(x);
8 for k = 0:N-1
9     for n = 0:N-1
10        X(n+1) = x(n+1)*exp(-j*2*pi*k*n/N);
11    endfor
12    Xk(k+1) = sum(X);
13 endfor
14 Xk
15
16
17 mag = abs(Xk);
18
19 figure(1)
20 subplot(2,1,1)
21 stem(0:N-1,mag, 'b', 'linewidth', 3);
22 legend ('jX_kj')
23 xlim([-0.5 3.5]);
24 ylim([-0.5 6.5]);
25 title('Magnitude and Phase of DFT')
26
27 phas = angle(Xk);
28 subplot(2,1,2)
29 stem(0:N-1,phas, 'r', 'linewidth', 3);
30 legend ('\angle Xk')
31 xlim([-0.4 3.4]);

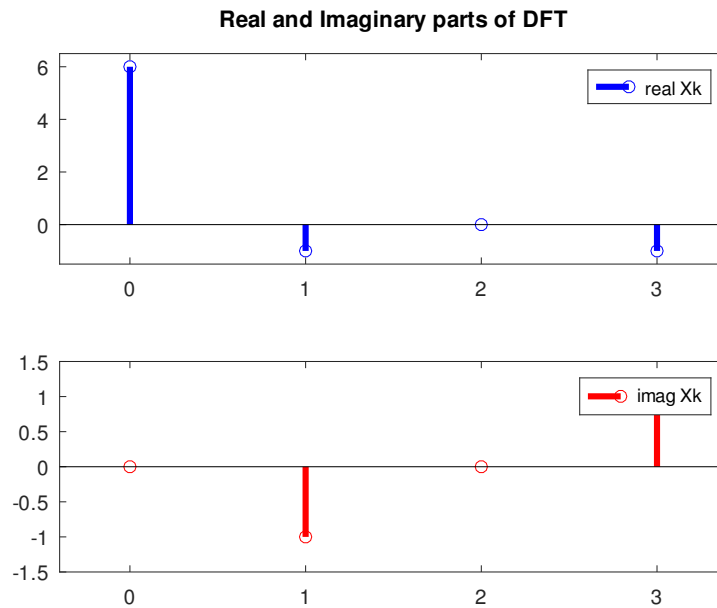
```

```

32 ylim([-3.5 3]);
33
34
35 re = real(Xk);
36 figure(2)
37 subplot(2,1,1)
38 stem(0:N-1,re, 'b', 'linewidth', 3);
39 xlim([-0.4 3.4]);
40 ylim([-1.5 6.5]);
41 legend('real Xk')
42 title('Real and Imaginary parts of DFT')
43
44 im = imag(Xk);
45 subplot(2,1,2)
46 stem(0:N-1,im, 'r', 'linewidth', 3);
47 xlim([-0.4 3.4]);
48 ylim([-1.5 1.5]);
49 legend('imag Xk')

```

Figure 3: Magnitude and Phase of  $X_k$

Figure 4: Real and Imaginary parts of  $X_k$ 

## 5 Lab 11: Lab Tasks

1. Compute and plot the DTFTs of the rectangular pulse

$$x[n] = \begin{cases} 1, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

2. Compute the DFT  $X_k$  of the sequence  $x[n] = [1, -2, 2, 1]$ ,  $0 \leq n \leq 3$ . Plot the magnitude, the phase, the real part, and the imaginary part of  $X_k$ .

## References

- [PV10] Alex Palamides and Anastasia Veloni. *Signals and systems laboratory with MATLAB*. CRC press, 2010.